Three Coordinate Quasitraces Refute Conjecture 00000000918

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October 30, 2026

Abstract

The printed assertion that the quasitrace simplex of every nuclear algebra has dimension one fails already for the finite-dimensional commutative C*-algebra $A = \mathbb{C}^3$ with its supremum norm. Its three coordinate evaluations are positive normalized weakly continuous traces with genuine compatible matrix quasitrace extensions. They are affinely independent over $\mathbb{R}$. The full quasitrace space containing them therefore cannot be one-dimensional. We formalize an exact completely positive approximation witness, the quasitrace and matrix-extension laws, the full-space dictionary, and the closed negation of the necessary universal dimension clause.

1 The Clause Being Refuted

Both source languages assert a simplex structure for the quasitrace space of a nuclear algebra, identify its extreme points with factor traces, and require that the same simplex have dimension one and a closed extreme-point set. Neither imposes simplicity, factoriality of the algebra, or restriction to a selected one-parameter family. We refute only the universal *dimension-one* requirement. We do not assert that the other clauses are false.

We use the usual nuclear C*-algebra and complex-valued quasitrace setting. In particular, dimension of a trace simplex means the real dimension of its affine span, not the dimension of an individual extreme-point factor. The usual normalized convention is $\tau(1) = 1$. Omitting normalization does not remove the three witnesses below.

Theorem 1. *There exists a commutative, finite-dimensional, unital nuclear C*-algebra whose full normalized weakly continuous* 2*-quasitrace space has real affine dimension at least two. The corresponding space without normalization also does not have affine dimension one.*

Since this is a genuine nuclear algebra, a universal assertion for nuclear algebras must apply to it. Restricting the printed clause to finite-dimensional commutative nuclear algebras is already enough to obtain a false statement.

2 The Actual Algebra and Nuclearity Witness

Let A be the functions on $\{0, 1, 2\}$ with values in $\mathbb{C}$, with pointwise operations, coordinate conjugation, unit $(1, 1, 1)$, and

$$\|a\| = \max_{0 \le i < 3} |a_i|.$$

This is the ordinary supremum-norm C*-algebra $\mathbb{C}^3$, not Euclidean space with its ℓ^2 norm. Its complex dimension is three.

Use the standard completely positive approximation property (CPAP) characterization of nuclearity: for every finite $E \subset A$ and $\varepsilon > 0$, there is a finite-dimensional C*-algebra F and completely positive contractions $\phi : A \to F$, $\psi : F \to A$ such that

$$\|\psi(\phi(a)) - a\| < \varepsilon \qquad (a \in E).$$

Take $F = A$ and $\phi = \psi = \mathrm{id}_A$. Every matrix amplification of the identity is the actual identity, hence preserves positivity; both maps are contractions and every error is zero. Thus the full finite-set and positive-error quantifiers hold, rather than being assumed from a label attached to A. The standard CPAP/nuclearity dictionary is the mathematical convention used here [3, 2]. The Lean result proves CPAP directly; it does not claim a new formal proof of its tensor-norm equivalence.

3 Genuine Quasitraces and Matrix Extensions

For $i \in \{0, 1, 2\}$ put $\tau_i(a) = a_i$. This is an actual continuous complex-linear functional, with $\tau_i(1) = 1$ and $|\tau_i(a)| \le \|a\|$. Moreover,

$$\tau_i(ab) = \tau_i(ba), \qquad \tau_i(a^*a) = \tau_i(aa^*) = |a_i|^2 \ge 0, \qquad \tau_i(a^*) = \overline{\tau_i(a)}.$$

Thus each is a positive normalized trace.

For clarity, a standard 1-quasitrace has positive cyclic star-square values, real values on selfadjoint elements, compatibility with the selfadjoint real/imaginary decomposition, and linearity on abelian C*-subalgebras. A 2-quasitrace additionally has a compatible 1-quasitrace on $M_2(A)$ [1]. Every positive complex-linear star-preserving trace satisfies these laws. In particular, additivity gives the source's subadditivity with equality on the selfadjoint/positive part, where the order inequality is defined.

Here the compatible extension is explicit. For $X \in M_2(A)$ set

$$Q_i(X) = \mathrm{Tr}_2\big[(\tau_i(X_{rs}))_{r,s=0,1}\big] = \tau_i(X_{00}) + \tau_i(X_{11}).$$

This uses the *unnormalized* ordinary matrix trace. It is complex-linear, star-preserving and tracial, and

$$Q_i(X^*X) = \sum_{r,s=0}^{1} |(X_{rs})_i|^2 \ge 0.$$

It therefore satisfies the full 1-quasitrace laws on the actual matrix C*-algebra. For the standard corner embedding,

$$Q_i\begin{pmatrix} a & 0 \\ 0 & 0 \end{pmatrix} = \tau_i(a), \qquad Q_i(1_2) = 2.$$

The value two on the matrix identity is correct; imposing normalization one on this extension would break corner compatibility. Thus the τ_i are genuine normalized 2-quasitraces, not three functions merely named quasitraces.

For $\sigma(A, A^*)$, every continuous linear functional is weakly continuous by the definition of the weak topology. Lean checks this on its actual `WeakSpace` for each τ_i. In finite dimension, the ordinary weak and norm topology descriptions do not reduce these three distinct functionals to a one-parameter family. The global theorem that quasitraces on exact algebras are traces is not needed.

4 The Full Space and Its Affine Dimension

Let p_j be the coordinate projection with value one at j and zero elsewhere. These are actual elements of A, and $\tau_i(p_j) = \delta_{ij}$. If

$$r(\tau_1 - \tau_0) + s(\tau_2 - \tau_0) = 0 \qquad (r, s \in \mathbb{R}),$$

evaluation at p_1 and p_2 gives $r = 0$ and $s = 0$. Hence the three coordinate traces are affinely independent over $\mathbb{R}$.

This is an obstruction for the *entire* space. Let S be any set in the real continuous dual containing all three τ_i. Its affine-span direction contains the two independent differences, so its dimension is at

least two. Taking S to be the full normalized weak 2-quasitrace space proves Theorem 1. Taking the full space without normalization proves the second assertion. The set S is not defined to be the three displayed points or their convex hull.

There is no hidden restriction to preselected linear functionals: on this commutative algebra, the quasitrace axiom of linearity on the whole closed abelian subalgebra A gives an actual complex-linear functional. Finite dimension makes it continuous. Lean proves its unique representation in $A \to_L \mathbb{C}$, and identifies the full unbundled normalized space with the image of that set under an injective real-linear coefficient map. Thus the dual is the natural affine ambient space for the original functionals. Its real finite-dimensionality is established before using `finrank`; no infinite-dimensional `finrank=0` convention is exploited.

We only require dimension at least two, not an exact dimension computation, an affine-homeomorphism classification, or a theorem about all nuclear algebras. The factor-trace and closed-extreme-set clauses are not attacked.

5 Formal Scope and Evidence Boundaries

The five Lean modules use Lean 4.33.1 and fixed Mathlib revision `0df444a360eaa60ab8c11dca51a86af692955474`. They separate the exact CPAP witness, genuine Pi-norm algebra and continuous-dual geometry, actual operator-norm `CStarMatrix` extension, quasitrace axioms and full-space dictionary, and final closed negation `source_necessary_clause_false`. The matrix base order is the canonical spectral order of the same C*-algebra, not a replacement of its norm by the raw entrywise matrix norm.

All 68 public theorem types and transitive axiom footprints have actual author-side compiler output. Allowed logical axioms are only `propext`, `Classical.choice`, and `Quot.sound`. Failed compilation attempts, including an initial 60-second CPAP timeout, are retained separately and are not counted as successes. A small exact rational-complex diagnostic checks coordinate arithmetic and a nonzero affine minor; those finite checks do not establish nuclearity, general quasitrace laws, weak continuity or the all-input Lean statements.

Original source blob: `dadc2ef65f6a57feb3def12febafd66299b1f918`, checked in both languages. This research manuscript and author-side module checks do not constitute independent frozen-package/PDF review, portable whole-package reproduction, current duplicate clearance or upstream acceptance. Existing compiler and official dependency caches remain trust inputs; no full-dependency-clean, empty-machine, Linux or hosted-CI result is claimed.

References

[1] U. Haagerup, *Quasitraces on exact C*-algebras are traces*, C. R. Math. Rep. Acad. Sci. Canada 36 (2–3) (2014), 67–92; arXiv:1403.7653. Introduction: quasitrace and compatible matrix-extension conventions. The global theorem is not used as an assumption.

[2] J. Castillejos, S. Evington, A. Tikuisis, S. White and W. Winter, *Nuclear dimension of simple C*-algebras*, Inventiones Mathematicae (2021), DOI 10.1007/s00222-020-01013-1, Section 1.1.

[3] M.-D. Choi and E. G. Effros, *Nuclear C*-algebras and the approximation property*, American Journal of Mathematics 100 (1) (1978), 61–79. Foundational bibliographic reference, not a claim to have independently audited the entire original equivalence proof.